CISB5123 (Text Analytics)

Sentiment Analysis of UNITEN Student Feedback on Campus Laundry Service

Project - Part 1

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1.0 Introduction

In university residential settings, facility services significantly affect students' daily living experience. At UNITEN residential colleges (Murni, Ilmu, Cendekiawan, Amanah), students frequently express dissatisfaction regarding the campus laundry service. Common issues include:

- Machines deducting payment without operating
- Insufficient machine availability during peak hours
- Poor drying performance
- Inaccurate timers
- Unhygienic laundry areas

Currently, these complaints are shared informally through verbal conversations or social media platforms such as Telegram confession channels. However, no structured system exists to systematically analyze these feedback patterns and support data-driven improvements.

2.0 Project Scope

2.1 Problem Addressed:

This project addresses the lack of structured analysis of student feedback regarding the campus laundry service. Although students frequently express complaints, there is no systematic evaluation of overall sentiment or identification of recurring problem themes.

Without structured analysis:

- Management cannot prioritize key issues.
- Service quality improvements are not evidence-based.
- Student satisfaction trends remain unknown.

2.2 Insights to extract:

The project aims to extract the following insights:

- Overall sentiment of student feedback (positive, negative, neutral).
- Frequently mentioned complaints and recurring issues.
- Key improvement factors affecting student satisfaction.

These insights will help determine whether dissatisfaction is dominant and what specific areas require immediate attention.

2.3 Data Sources/audiences:

The primary data source is a Google Forms survey distributed to UNITEN residential students. The survey includes rating-based questions and open-ended text responses.

If necessary, additional publicly available Telegram confession posts related to the laundry service may be used.

Target Audience:

- UNITEN hostel management
- Laundry service operators
- Residential students

The analysis aims to support data-driven decision-making for service improvement.

2.4 Ethical considerations:

The project considers the following ethical aspects:

- **Data Privacy**: No personal identifiers (name, student ID, contact information) are collected.
- **Consent**: Participation in the survey is voluntary, and respondents are informed about the purpose of the study.
- **Bias Awareness**: Feedback is interpreted carefully to reduce misclassification and overgeneralization.

These considerations ensure responsible and ethical use of textual data

3.0 Collect and describe your dataset.

3.1 Data taken from:

The data for this project will mainly be collected through online surveys via Google Forms, distributed to social media channels used by UNITEN students (Whatsapp Group and Telegram). The survey will include ratings, and open-ended questions to allow students to express their opinions freely.

3.2 Dataset size:

- Estimated number of records: 50 - 100 respondents
- File Format: Google form responses will be exported as CSV file
- Source: Google Form
- Data type: Structured survey data with open-ended textual responses
- Language: Primarily English, with possible Malay-English mixed responses

3.3 Potential data limitations:

Several limitations may affect the dataset:

1. Limited data volume if response rate is low.
2. Response bias, as dissatisfied students may be more likely to respond.
3. Informal and unstructured text containing slang, mixed languages, spelling errors, and emojis.
4. Convenience sampling may not fully represent all residential students.

4.0 Perform text preprocessing.

4.1 Steps applied

For AI to analyze the student's sentiment on UNITEN on-campus laundry, we are required to perform text cleaning. The below includes the steps performed:

1) **Removal of weird, encoded characters** – Removed non-text symbols that do not contribute to semantic meaning.
2) **Lowercasing** – All text converted to lowercase for consistency.
3) **Replace Slangs** – Slangs are replaced with its original meaning
4) **Translation of Malay** – Malay words are converted to English for accuracy
5) **Removal of punctuation** – Text are standardized to remove any irrelevant characters.
6) **Replace Contractions** – Short forms texts are replaced with its long form
7) **Spelling Errors Corrected** – Texts not autocorrected by Speller is to be corrected manually
8) **Removal of numbers** – Numbers included are removed as it does not contribute to assessing student' sentiment
9) **Correct Spelling** – Spelling automatically corrected through Speller library
10) **Stopword removal** – Removed common words such as "the", "is", and "and" that do not carry significant meaning.
11) **Lemmatization** – Reduce words to their base form to improve consistency.
12) **Tokenization** – Split text into individual words (tokens) for analysis

4.2 Why the steps were necessary

The survey responses contain informal language, mixed expressions, and inconsistent formatting. Preprocessing is necessary to:

- Standardize textual data
- Remove irrelevant noise
- Improve sentiment analysis accuracy
- Enhance model performance in later stages

Without preprocessing, the analysis may produce inaccurate or misleading results.

4.3 Challenges faced during preprocessing

Several challenges were encountered during the preprocessing stage:

- **Short and vague comments**: which limited contextual understanding and sentiment interpretation.
- **Informal grammar and spelling inconsistencies**: making text normalization difficult.
- **Use of slang expressions**: which may not be recognized by standard preprocessing tools.
- **Same words but in different tenses**: requiring lemmatization to standardize word forms (e.g., “run”, “running”, “ran”).
- **Words in Malay**: requires a Malay dictionary to be created.